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CONNECTION DESIGN ADVISOR

Project: Quick Analysis

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Model: STRUCTMIND CORE · PRO

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Acknowledged. As STRUCTMIND AI, I will provide optimal connection recommendations for your 3-storey SMF + OCBF mixed-use system. The analysis below is organized by frame type and connection location, referencing AISC 360-22 and relevant seismic provisions in AISC 341-22.

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1. Special Moment Frame (SMF) Beam-to-Column Connections

These connections are designed to be ductile and dissipate seismic energy through controlled inelastic deformation (plastic hinging) in the beams. They must be prequalified per AISC 358.

• 1. Recommended Connection Type: Reduced Beam Section (RBS) or "Dogbone"

• This is a prequalified, fully-restrained (FR) moment connection. A portion of the beam flanges is trimmed in a curved "dogbone" shape to create a fusible link, forcing the plastic hinge to form away from the column face. This protects the critical beam-to-column weld.

• 2. Governing Limit States (per AISC 360-22 Chapter J and AISC 341-22 Chapter E):

• Primary Design Goal (Ductility): Flexural yielding of the beam gross section at the reduced section (this is the intended behavior, not a failure state).

• Beam: Shear yielding/rupture of the beam web; Local flange buckling and local web buckling at the hinge location.

• Connection Elements:

• Weld: Strength of the CJP groove weld connecting the beam flange to the column flange. Strength of the fillet or CJP weld for the beam web-to-shear tab connection. (Note: These are designated "Demand Critical" welds per AWS D1.8).

• Bolts (Web Connection): Bolt shear rupture; Bolt bearing on beam web and shear tab.

• Column:

• Panel Zone Shear Yielding (AISC 341 E3.6g): This is often a primary governing limit state. The column web panel must be strong and stiff enough to transfer moments. Doubler plates may be required.

• Strong Column-Weak Beam (AISC 341 E3.4a): The sum of column flexural strengths must exceed the sum of beam flexural strengths at the joint to prevent soft-story mechanisms.

• Continuity Plates (AISC 360 J10.8): Required to prevent column flange local bending and web local crippling.

• 3. Bolt / Weld Sizing Guidance:

• Bolts: Use high-strength ASTM F3125 Gr A325 or A490 bolts for the web connection. Design for the shear corresponding to the probable maximum moment (Mpr) of the beam at the plastic hinge.

• Welds: Beam flanges must be connected to the column flange with Complete Joint Penetration (CJP) groove welds. These are Demand Critical and require rigorous inspection (NDE/NDT).

• 4. Constructability Score & Remarks:

• Score: 7/10

• Remarks: The RBS cut is a standard CNC plasma or oxy-fuel cutting operation in the fabrication shop. However, the CJP field or shop welding requires highly skilled labor and extensive QA/QC. Erection requires careful alignment. Overall, it's a very common but technically demanding connection.

• 5. Alternate Solutions:

• Bolted Flange Plate (BFP) Connection: Plates are bolted to the top and bottom beam flanges and welded to the column. Reduces field welding but requires tight tolerances.

• Bolted End Plate (BFP) Connection: A stiffened end plate is shop-welded to the beam and then field-bolted to the column face. Very fast erection but requires heavy plates and large bolts.

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2. OCBF & Gravity Beam-to-Column Connections

These are designed as simple (pinned) connections, assumed to transfer shear only and allow for end rotation.

• 1. Recommended Connection Type: Single Plate Shear Connection (Shear Tab)

• A single plate is shop-welded to the column and field-bolted to the beam web. This is the modern industry standard due to its economy and ease of erection.

• 2. Governing Limit States (per AISC 360-22, J4 & J5):

• Bolt Shear Rupture

• Bolt Bearing on the beam web and the plate

• Shear Yielding of the plate (gross section)

• Shear Rupture of the plate (net section)

• Flexural Yielding of the plate (at the weld line)

• Block Shear Rupture on the beam web

• Block Shear Rupture on the plate

• Weld Strength (plate-to-column)

• 3. Bolt / Weld Sizing Guidance:

• Bolts: ASTM F3125 Gr A325 in a bearing-type connection is standard. Size based on the beam's shear reaction. Use short-slotted holes in the plate to provide erection tolerance.

• Welds: Use standard fillet welds. Size for the shear reaction, ensuring it meets AISC minimums based on plate thickness.

• 4. Constructability Score & Remarks:

• Score: 9/10

• Remarks: Highly constructible. Fabrication is simple (one plate, one weld pass). Erection is fast and forgiving due to the single line of bolts and potential for slotted holes.

• 5. Alternate Solutions:

• Double Angle Connection: Two angles either bolted or welded to the column and bolted to the beam web. A classic, reliable connection, but involves more pieces and labor than a shear tab.

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3. OCBF Brace-to-Gusset Connection

This connection transfers the large axial forces from the brace (e.g., HSS member) into the frame at the beam/column intersection.

• 1. Recommended Connection Type: Gusset Plate Connection (Uniform Force Method)

• A gusset plate is welded to the beam and/or column. The HSS brace is typically shop-welded directly to the gusset plate by slotting the HSS over the plate.

• 2. Governing Limit States (per AISC 360-22, J4 and AISC 341-22, F1):

• Brace: Tensile yielding on gross area; Tensile rupture on net effective area.

• Gusset Plate:

• Tensile Yielding on the Whitmore effective width section.

• Block Shear Rupture at the brace interface.

• Compressive Buckling of the gusset plate.

• Interface:

• Strength of the weld between brace and gusset.

• Strength of welds connecting the gusset to the beam/column flanges.

• OCBF Specific (AISC 341 F1.6): The connection must be designed to accommodate the expected inelastic deformation of the brace, including providing clearance for out-of-plane buckling. A small hinge zone is often detailed.

• 3. Bolt / Weld Sizing Guidance:

• Welds: Size fillet welds to develop the required strength of the brace (tensile yielding or expected buckling strength). Ensure sufficient length along the gusset edges.

• Gusset Thickness: Governed by Whitmore yielding or buckling criteria. Typically between 1/2" and 1" thick for a low-rise building.

• 4. Constructability Score & Remarks:

• Score: 8/10

• Remarks: Fabrication is straightforward but requires precise layout to ensure work points align. Shop-welding the brace to the gusset is highly efficient. Field erection involves welding the gusset assembly to the main frame or bolting the beam with the attached gusset into place.

• 5. Alternate Solutions:

• Bolted Brace Connection: The HSS brace is welded to a tee or a paddle plate in the shop, which is then field-bolted to the gusset. This eliminates field welding but adds material and fabrication steps.

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4. Column Base Plates

• 1. Recommended Connection Type:

• SMF Columns: Fixed Base Plate. Designed to resist axial load, shear, and significant bending moment. Typically uses 8 anchor rods (4 inside, 4 outside flanges) to create a moment couple.

• OCBF/Gravity Columns: Pinned Base Plate. Designed to resist axial load and shear only. Typically uses 4 anchor rods located inside the column flanges.

• 2. Governing Limit States (per AISC 360-22, J8 and AISC Design Guide 1):

• Concrete: Concrete Bearing Strength under the plate (AISC J8).

• Base Plate: Plate Bending (flexural yielding) at critical sections around the column and stiffeners.

• Anchors (per ACI 318): Anchor Rod Tensile Strength; Concrete Breakout Strength (tension & shear); Anchor Rod Shear Strength; Concrete Pryout Strength.

• Weld: Strength of the weld connecting the column to the base plate.

• 3. Bolt / Weld Sizing Guidance:

• Anchors: Use ASTM F1554 Gr 36, 55, or 105 anchor rods. Sizing is an iterative process governed by ACI 318, balancing rod diameter, embedment depth, and edge distances.

• Plate Thickness: Driven by the bearing pressure and the resulting bending moment in the plate. For fixed bases, thickness can become substantial (2" to 4"+). Stiffeners can help reduce thickness.

• Welds: Fillet welds are common. For high-moment SMF bases, CJP or PJP groove welds at the flanges may be necessary.

• 4. Constructability Score & Remarks:

• Score: 7/10

• Remarks: Success is highly dependent on coordination with the concrete trade. Anchor bolt placement using a template is critical. Leveling nuts and a proper grouting procedure are essential for achieving the correct elevation and full bearing. Fixed bases are significantly more complex to install than pinned bases.

• 5. Alternate Solutions:

• Embedded Column Pier: The column is embedded directly into the concrete foundation. Provides excellent fixity but complicates foundation forming and waterproofing. Not typical for a 3-storey building.

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5. Column Splices

Splices are typically located every two stories (above the 2nd floor in this case) and away from the beam-column joint to avoid interfering with connection elements and high-moment regions.

• 1. Recommended Connection Type: Bolted Flange Plate Splice

• Plates are used on the outside of the flanges (and sometimes the web) to transfer the load across the joint. This is a field-bolted solution.

• 2. Governing Limit States (per AISC 360-22, J4 and AISC 341-22, D2.5):

• OCBF/Gravity Columns:

• Bolt Shear Rupture and Bearing.

• Gross Section Yielding and Net Section Rupture of the splice plates.

• Block Shear Rupture of the plates and column flanges.

• SMF Columns (AISC 341 D2.5b): The splice must be designed to develop the required flexural and shear strength of the column. The demands are significantly higher than the analysis forces. Splices must be located in the middle third of the column clear height.

• 3. Bolt / Weld Sizing Guidance:

• Bolts: Use high-strength ASTM F3125 Gr A325 or A490 bolts. For SMF columns, slip-critical connections are required to prevent any slippage under load reversal. For OCBF/Gravity columns, bearing-type connections are sufficient.

• Plates: Sized to meet the yielding and rupture limit states based on the forces being transferred.

• 4. Constructability Score & Remarks:

• Score: 8/10

• Remarks: Excellent for erection speed and safety. Erection aids (e.g., guide lugs) are often shop-welded to the lower column to help seat and align the upper column before bolting. Fabrication tolerances are important for proper fit-up.

• 5. Alternate Solutions:

• Directly Welded Splice: Uses CJP groove welds to directly connect the upper and lower column sections. Provides a seamless aesthetic but is extremely expensive, slow, and requires extensive field NDT inspection. It is generally avoided in commercial construction.